

CS 506: Introduction to Quantum Computing

Scribe Notes: Linear Operators, Reversibility, Gates, and Composite Systems

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1 Linear Operators and the NOT Gate

The NOT gate maps the two basis states as

$$0 \mapsto 1, \quad 1 \mapsto 0.$$

In vector form,

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The corresponding linear operator is

$$\text{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

More generally,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = \begin{pmatrix} p_1 \\ p_0 \end{pmatrix}.$$

This makes sense intuitively: NOT just swaps the two amplitudes/probabilities, which is exactly what flipping a bit should do.

2 Writing Operators in Dirac Notation

Consider the operator

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its action is

$$|0\rangle \mapsto |0\rangle, \quad |1\rangle \mapsto -|1\rangle.$$

The same operator can be written directly in Dirac notation as

$$|0\rangle\langle 0| - |1\rangle\langle 1|.$$

Indeed,

$$(|0\rangle\langle 0| - |1\rangle\langle 1|)|0\rangle = |0\rangle,$$

and

$$(|0\rangle\langle 0| - |1\rangle\langle 1|)|1\rangle = -|1\rangle.$$

Multiplying a column vector by a row vector gives a matrix – this is worth pausing on since it's easy to mix up with the more familiar row-times-column case. For example,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

while

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

So row $\times$ column gives an inner product (a scalar), while column $\times$ row gives an outer product (a matrix).

3 Reversibility

A gate G is reversible if, whenever it transforms an input x into an output y , there is an inverse gate G^{-1} that transforms y back into x . In other words, no information is lost in the process.

For the NOT gate,

$$\text{NOT}|0\rangle = |1\rangle, \quad \text{NOT}|1\rangle = |0\rangle.$$

Applying NOT again restores the original state, so

$$\text{NOT}^{-1} = \text{NOT}.$$

The NOT gate is therefore its own inverse.

Not every classical gate has this property. The ordinary AND gate has truth table

x	y	$x \wedge y$
0	0	0
0	1	0
1	0	0
1	1	1

so the output alone doesn't tell you which input pair produced it – three different inputs all map to 0. This means AND, as written, can't be reversible: information about the inputs is destroyed.

The fix is to keep the original inputs around and add an extra bit z to store the result:

$$(x, y, z) \mapsto (x, y, z \oplus (x \wedge y)).$$

Applying the same operation twice gives back the original z :

$$(z \oplus (x \wedge y)) \oplus (x \wedge y) = z \oplus 0 = z,$$

so this version is reversible even though ordinary AND wasn't.

4 Control-NOT (CNOT)

The control-NOT gate acts on a two-bit input. The control bit c is left unchanged, and the other bit gets XOR'd with the control:

$$(c, x) \mapsto (c, x \oplus c).$$

With basis ordering 00, 01, 10, 11, the matrix form is

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

This gives the input/output mappings

$$00 \mapsto 00, \quad 01 \mapsto 01, \quad 10 \mapsto 11, \quad 11 \mapsto 10.$$

As an example,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

which matches $10 \mapsto 11$ above.

5 Unitary Operators

Operators describing the time evolution of a closed quantum system are always unitary. This is a core postulate of quantum mechanics, not something derived from the gates themselves.

A useful example is

$$A = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

It is unitary because

$$A^\dagger A = I,$$

where $A^\dagger$ is the complex-conjugate transpose and I is the identity matrix.

Unitary operators preserve the norm of vectors and preserve inner products between vectors – which is exactly what you'd want, since probabilities need to stay normalized under evolution. In fact, any unitary matrix corresponds to a valid quantum gate; unitarity is the only real constraint.

6 Composite Systems and Tensor Products

It helps to first compare two individual probabilistic bits against a composite two-bit system. Using the labeling

$$00 \equiv 0, \quad 01 \equiv 1, \quad 10 \equiv 2, \quad 11 \equiv 3,$$

suppose

$$\begin{aligned}\Pr(X = 0) &= \frac{1}{3}, & \Pr(X = 1) &= \frac{2}{3}, \\ \Pr(Y = 0) &= \frac{1}{2}, & \Pr(Y = 1) &= \frac{1}{2}.\end{aligned}$$

If X and Y are independent, the composite probabilities multiply:

$$\Pr(00) = \frac{1}{6}, \quad \Pr(01) = \frac{1}{6}, \quad \Pr(10) = \frac{1}{3}, \quad \Pr(11) = \frac{1}{3}.$$

The quantum analogue uses tensor products instead of ordinary multiplication. Given two individual qubits

$$\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$

and

$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle,$$

their composite state is

$$\begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}.$$

Notice the amplitudes here are products of the individual amplitudes, mirroring how the classical probabilities multiplied above.

7 A Reversible Version of a Classical Gate

Consider a classical gate G with two-bit inputs

$$x = (x_1, x_2), \quad y = (y_1, y_2),$$

and a two-bit output $z = (z_1, z_2)$ defined by

$$z_1 = x_1 \wedge y_1, \quad z_2 = x_2 \vee y_2.$$

Since both AND and OR lose information about their inputs, this gate isn't reversible as written. Following the same trick as before, a reversible version keeps the input bits and adds two extra bits w_1, w_2 to absorb the outputs:

$$w_1 \oplus (x_1 \wedge y_1), \quad w_2 \oplus (x_2 \vee y_2).$$

This is the general pattern for reversibilizing any classical circuit – pair each output with an ancilla bit via XOR.

8 Dimension Two: Standard Basis and Hadamard Basis

The standard basis is

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so any qubit state can be written as

$$\begin{pmatrix} p \\ q \end{pmatrix} = p|0\rangle + q|1\rangle.$$

There's nothing special about this basis, though – the same state can equally well be described in the Hadamard basis:

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \\ |-\rangle &= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}. \end{aligned}$$

Adding and subtracting these gives

$$|+\rangle + |-\rangle = \sqrt{2}|0\rangle, \quad |+\rangle - |-\rangle = \sqrt{2}|1\rangle,$$

so we can solve for the standard basis in terms of the Hadamard basis:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle), \quad |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle).$$

Substituting back,

$$p|0\rangle + q|1\rangle = \frac{p}{\sqrt{2}}(|+\rangle + |-\rangle) + \frac{q}{\sqrt{2}}(|+\rangle - |-\rangle),$$

which simplifies to

$$\left(\frac{p}{\sqrt{2}} + \frac{q}{\sqrt{2}}\right)|+\rangle + \left(\frac{p}{\sqrt{2}} - \frac{q}{\sqrt{2}}\right)|-\rangle.$$

So the same physical state has completely different-looking coordinates depending on which basis you choose to express it in – a point worth keeping in mind going forward, since a lot of quantum algorithms exploit exactly this kind of basis change.